

MATHEMATICS – YEAR 10

Achievement Standard Guide



About this Achievement Standard Guide

This Achievement Standard Guide describes the expected knowledge, skills and understanding students should demonstrate for Mathematics in Year 10.

The Guides were developed using the achievement standards of Version 9.0 of the Australian Curriculum. The achievement standards assist teachers to monitor student learning and to make judgements about student progress and achievement.

The Guides set out the aspects of the achievement standard and describe indicators at five levels of attainment. They identify possible observable characteristics that could demonstrate the level of the student's understanding.

The Achievement Standard Guides were developed to:

- support teacher judgement when assessing student learning
- support planning for learning in all aspects of the achievement standard
- support the development of clear and specific learning intentions and success criteria.

The Achievement Standard Guides can be used to:

- ensure consistency of assessment
- support moderation of student work at year, school, network and state level
- support unit and lesson planning
- ensure teachers have sufficient evidence of learning to make defensible on-balance judgements about the quality of students' work against the achievement standards.

An effective assessment program will:

- Guide and support targeted teaching and learning
- Ensure students have opportunities to demonstrate depth and breadth of their learning in all aspects of the achievement standard
- Provide regular feedback to students about how they can improve in their learning
- Clarify future teaching and learning needs.

The following information supports teachers in these aspects of quality assessment practice.

Mathematics – Year 10 Achievement Standard

Strand	By the end of Year 10, students:
<u>Number</u>	recognise the effect of approximations of real numbers in repeated calculations
<u>Algebra</u>	- use mathematical modelling to solve problems involving growth and decay in financial and other applied situations, applying linear, quadratic and exponential functions as appropriate, and solve related equations, numerically and graphically - make and test conjectures involving functions and relations using digital tools - solve problems involving simultaneous linear equations and linear inequalities in 2 variables graphically and justify solutions
<u>Measurement</u>	- interpret and use logarithmic scales representing small or large quantities or change in applied contexts - solve measurement problems involving surface area and volume of composite objects - apply Pythagoras' theorem and trigonometry to solve practical problems involving right-angled triangles - identify the impact of measurement errors on the accuracy of results - use mathematical modelling to solve practical problems involving proportion and scaling, evaluating and modifying models, and reporting assumptions, methods and findings
<u>Space</u>	- use deductive reasoning, theorems and algorithms to solve spatial problems - interpret networks used to represent practical situations and describe connectedness
<u>Statistics</u>	- plan and conduct statistical investigations involving bivariate data - represent the distribution of data involving 2 variables, using tables and scatter plots, and comment on possible association - analyse inferences and conclusions in the media, noting potential sources of bias - compare the distribution of continuous numerical data, using various displays, and discuss distributions in terms of centre, spread, shape and outliers
<u>Probability</u>	- apply conditional probability to solve problems involving compound events - design and conduct simulations involving conditional probability, using digital tools.

How the Achievement Standard Guide is organised

The relevant achievement standard statement in italics.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	Achievement standard statement with bolded variations For example, students can: ...	Achievement standard statement For example, students can: ...	Achievement standard statement with bolded variations For example, students can: ...	

Content strand.

'Can do' statements associated with each attainment level, including possible observable characteristics as examples of what students can do.

Blank spaces or merged cells indicates a specific skill that can either be achieved or not.

Merged cell indicates that attainment is possible beyond 'At Standard' but not readily discernible between 'Above' or 'Well Above'

Number

Students recognise the effect of approximations of real numbers in repeated calculations.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Number	With support students recognise the effect of approximations of real numbers in repeated calculations For example, students can: With support recognise that rounding off early in repeated calculations will result in a less accurate answer than not rounding off until the end.	Students recognise the effect of approximations of real numbers in repeated calculations For example, students can: - Show the impact on the answer, when rounding numbers early in the calculation compared to not rounding until the end when solving real-world problems. - Eg When calculating volumes or surface areas of cylinders using π compared with approximations such as 3.14 or $\frac{22}{7}$ - Eg when rounding a trigonometric value such as $\tan(30^\circ)$ to one decimal place at the start of a calculation instead of waiting until the final answer before rounding.	Students recognise the effect of approximations of real numbers in repeated calculations and consistently plan for the appropriate level of accuracy in each context For example, students can: Consistently and independently consider the level of accuracy required when solving real-world problems and adjust numbers accordingly (eg solving real-world problems involving measurement, finance or exponential growth and decay).	

Algebra

Students use mathematical modelling to solve problems involving growth and decay in financial and other applied situations, applying linear, quadratic and exponential functions as appropriate, and solve related equations, numerically and graphically.

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Algebra	With support, they use mathematical modelling to solve problems involving growth and decay in financial and other applied situations, applying linear, quadratic and exponential functions as appropriate, and solve related equations, numerically and graphically. For example, students can: - With support use a mathematical modelling approach that allows them to: - Translate a real-world problem into maths - Use maths to solve the problem - Interpret and communicate the solution back in the real-world context with some - justification - evaluation and modification - When using linear, quadratic or exponential rules to model financial and other applied contexts: - Find the co-ordinate pairs for any points on a supplied linear graph, parabola or exponential graph. - Use technology to enter different types of given algebra rules to produce graphs, eg, by using Desmos - Recognise whether the rule is linear, quadratic or exponential	They use mathematical modelling to solve problems involving growth and decay in financial and other applied situations, applying linear, quadratic and exponential functions as appropriate, and solve related equations, numerically and graphically. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: - Translating the problem into maths - Using maths to solve the problem - Interpreting and communicating the solution back in the real-world context with: - justification - evaluation and modification and refinement. - When using linear, quadratic or exponential rules to model financial and other applied contexts: - Find the co-ordinate pairs for any points on a supplied linear graph, parabola or exponential graph. - Sketch a line of best fit by eye (ie without using a defined mathematical process or technological tool) to find coordinate pairs for points	They use mathematical modelling to solve problems involving growth and decay in financial and other applied situations, applying linear, quadratic and exponential functions as appropriate, and solve related equations numerically, graphically and algebraically. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: - Translating the problem into maths - Using maths to solve the problem - Interpreting and communicating the solution back in the real-world context with: - justification - evaluation, modification and refinement, providing valid reasoning. (The mathematics becomes increasingly difficult, and the evaluation and justification become more sophisticated). - When using linear, quadratic or exponential rules to model financial and other applied contexts:	They use mathematical modelling to solve problems involving growth and decay in financial and other applied situations, applying linear, quadratic and exponential functions as appropriate, and solve related equations, numerically, graphically and algebraically. They use a wider variety of algebraic techniques, convert fluently between different quadratic forms and use evidence of comparison to evaluate chosen models. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: - Translating the problem into maths - Using maths to solve the problem - Interpret and communicate the solution back in the real-world context - Justification, including detailed reasoning, identifying the strength of the supporting evidence. - evaluation includes checking accuracy, identification of assumptions, suggestions with valid reasoning to improve the quality of the model

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	(or none of these) from a graph. - Substitute given x values to complete a table of values when given a rule to model a real-world situation and be aware of whether the data points generated by the rule are close to the real-world data points or not. - Locate the y intercept and x intercepts on a graph and give co-ordinates.	which are in between or “outside” the original points (for linear graphs, parabolas and exponential graphs). - Use a supplied linear, quadratic or exponential rule to create a table of values and then draw a graph with correctly scaled axes on graph paper. - Generate a table of values that matches a linear graph, parabola or exponential graph. - Evaluate whether a linear, quadratic or exponential relationship (or none of these) is appropriate to model a real-world situation based on the shape of plotted real-world data points. - Use a technological tool (eg Excel, Desmos, Geogebra) to determine a line of best fit (linear, quadratic, exponential) for data in a real-world problem and evaluate whether this rule seems to be predicting coordinates for points which are close enough to the actual real-world data or not. - Interpret and communicate mathematical solutions using language (and other representations) that match the original context of the problem.	- Factorise non-monic quadratic rules where there are not too many options for factors of a and c, (eg quadratic rules of the form: $y = ax^2 + bx + c$ or $y = ax^2 + bx - c$ or $y = ax^2 - bx + c$ or $y = ax^2 - bx - c$ - where a single digit prime and c is a whole number with no more than two pairs of factors) then use the factorised form to algebraically determine values of x which make $y = 0$ and hence determine the x – intercepts of the graph. - Factorise ‘special’ monic quadratic relationships using a difference of two squares method or perfect square method and hence determine the values of x which make $y = 0$ and then determine the x – intercept(s) of the associated graphs. - When using an exponential rule of the form $y = ab^x$ to model a real-world situation, efficiently find unknown values of: y when values of a, b and x are known a when values of y, b and x are known b when values of y, a and x are known	(The mathematics increases in difficulty, and the evaluation and justification are focussed upon refinement). - When using linear, quadratic or exponential rules to model financial and other applied contexts: - Factorise non-monic quadratic rules of the forms $y = ax^2 + bx + c$ or $y = ax^2 + bx - c$ $y = ax^2 - bx + c$ or $y = ax^2 - bx - c$ - where there are quite a few options for factors of a and c (and a might be negative) then use the factorised form to algebraically determine values of x which make $y = 0$ and hence determine the x – intercepts of the graph. - Use a combination of factorising techniques (eg common factor then: trinomial, difference of two squares or perfect square) to factorise a quadratic rule then use the factorised form to algebraically determine values of x which make $y = 0$ and hence determine the x – intercepts of the graph. - Complete the square for a monic quadratic or a quadratic where the coefficient of x^2 is -1 and hence determine the turning point of the graph. Justify why the “completed the square form” shows a minimum

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			– using substitution, order of operations and inverse operations, and find unknown values of x when values of a, b and y are known using a guess, check and improve strategy.	or maximum value of a quadratic relationship. – Move fluently between expanded form, factorised form and “completed the square form” (for quadratic rules where the coefficient of x^2 is 1 or -1) to determine the y – intercept, x – intercepts and turning point for graphs of quadratic relationships. – Evaluate a chosen model for a real-world situation by using evidence of comparison, eg – In quadratic modelling, students show that the values of a, b and c they have chosen in their rule of the form $y = ax^2 + bx + c$ seem to be the best possible values by changing a then b then c in their technological tool (eg Excel, Desmos) and showing that the changed models do not fit the actual real-world data as well as the original model. – In exponential modelling, students show that the values of a and b they have chosen in their rule of the form $y = ab^x$ seem to be the best possible values by changing a then b in their technological tool (eg Excel, Desmos) and showing that the changed models do not fit the actual real-world data as well as the original model (eg in compound interest or depreciation, or in population increase or decline).

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Algebra	With support, students make and test conjectures involving functions and relations using digital tools. For example, students can: • Make conjectures (about linear, quadratic and exponential functions) involving one changing parameter when guided by the teacher on how to make testable conjectures. • Follow supplied processes to: – systematically test conjectures using digital tools – organise evidence so that trends and patterns become clearer – report on conjectures that have been made and tested. • Interpret the evidence and decide if the conjecture is true when guided by the teacher.	Students make and test conjectures involving functions and relations using digital tools. For example, students can: • Make conjectures (about linear, quadratic and exponential functions) which are testable using digital tools such as Excel, Desmos or Geogebra. At standard conjectures involve one changing parameter in a function and could be about the impact of the parameter on: – Graph shape for all three types of functions (with quadratics in expanded form) – Location of x and y intercepts, (eg further left, further right, higher, lower or in some cases students might be able to hypothesise about specific values) for all three types of functions (with quadratics in factorised and expanded form) – Location of intersection points for pairs of linear functions, (eg further left, further right, higher, lower) – When the graph will be above or below the x axis for all three types of functions – When a graph will be above or below a certain value for all three types of functions. • Test conjectures using digital tools: – systematically by only changing one parameter at a time and collecting enough evidence	Students make and test conjectures involving functions and relations using digital tools by predicting and verifying the impact of changing one parameter at a time and testing the impact of changing two parameters. For example, students can: • Use what they have learnt from testing at standard conjectures to predict what will happen to: – the shape of a linear, quadratic or exponential graph if any of the parameters are changed (one at a time) then use a technological tool to demonstrate that their prediction is true – the position of the intersection point of two linear relationships if any of the parameters are changed (one at a time) in either of the relationships then use a technological tool to demonstrate that their prediction is true. • Make, test and report on conjectures where two parameters are changed, eg – The impact on the shape of a linear, quadratic or exponential relationship if two parameters are changed – The impact on the location of the x and y intercepts for	Students make and test conjectures involving functions and relations using digital tools by predicting and verifying the impact of changing two parameters and testing the impact of changing three parameters. For example, students can: • Use what they have learnt from testing above standard conjectures to predict what will happen to: – the shape of a linear, quadratic or exponential graph if any two of the parameters are changed then use a technological tool to demonstrate that their prediction is true – the position of the intersection points of a linear function with: another linear function, quadratic function or exponential function, if any two of the parameters are changed then use a technological tool to demonstrate that their prediction is true (N.B. the two changed parameters might be in the same function or in different functions). • Make, test and report on conjectures about the shapes of graphs of quadratic functions where three parameters are changed. This includes investigating quadratic functions in: – Expanded form – Factorised form

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		about the impact of changing that parameter before moving onto the next parameter using the same process – by gathering enough evidence to indicate whether the conjecture is likely to be true (the evidence might not be totally complete), ie the student has enough evidence to decide whether the conjecture is true or not and they don't have to ask the teacher for the answer. • Accurately report on conjectures they have made and tested by: – Describing the conjecture – Describing the method used to test the conjecture – Presenting the evidence – Summarising their conclusion with a clear, logical statement that aligns with the evidence.	all three types of functions if two parameters are changed (with quadratics in factorised and expanded form) – The impact on the intersection points of a linear function with: another linear function, a quadratic function or an exponential function, if two parameters are changed (N.B. the two changed parameters might be in the same function or in different functions) – The impact on when the graph of a linear, quadratic or exponential function will be above or below the x axis if two parameters are changed. • Test conjectures using digital tools by gathering evidence that is “comprehensive” (ie they might not have included every single example but the evidence is complete enough that there is virtually no risk of finding an example which contradicts the evidence).	– Turning point form • then use what they have learnt to work out the quadratic function (without using a digital tool) when given a graph with the intercepts and turning point labelled. The functions should be expressed in factorised form and expanded form (and turning point form for quadratic rules where the coefficient of x^2 is 1 or -1). • Make, test and report on conjectures about intersection points of any two different types of functions (linear, quadratic, exponential) where three parameters are changed.

Students solve problems involving simultaneous linear equations and linear inequalities in 2 variables graphically and justify solutions.

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Algebra	With support, they solve problems involving simultaneous linear equations and linear inequalities in 2 variables graphically and justify solutions. For example, students can: - Identify and state the point of intersection for two given linear graphs. - Follow a supplied process to: - Graphically solve pairs of simultaneous equations such as $y = 2x + 25$ and $y = 7x - 10$, with and without technology, and algebraically show that the coordinate pair for the intersection point is a solution of both rules by substituting and calculating. - Use the two graphs to state the solutions to the related inequalities, ie for which values of x is $2x + 25 > 7x - 10$ and for which values of x is $2x + 25 < 7x - 10$? - Follow a supplied process to: - Graphically solve real-world worded problems involving simultaneous linear relationships with and without technology, eg one plumber charges \$150 up front plus \$65 per hour and another plumber charges \$40 up front plus \$90 per hour so find the number of hours which results in the same	They solve problems involving simultaneous linear equations and linear inequalities in 2 variables graphically and justify solutions. For example, students can: - Graphically solve pairs of simultaneous equations such as $y = 2x + 25$ and $y = 7x - 10$, with and without technology, and algebraically show that the coordinate pair for the intersection point is a solution of both rules by substituting and calculating. Then use the two graphs to state the solutions to the related inequalities, ie for which values of x is $2x + 25 > 7x - 10$ and for which values of x is $2x + 25 < 7x - 10$? - Graphically solve real-world worded problems involving simultaneous linear relationships with and without technology, eg one plumber charges \$150 up front plus \$65 per hour and another plumber charges \$40 up front plus \$90 per hour so find the number of hours which results in the same charge for each plumber. (Students don't necessarily have to write the algebra rule from the words to generate the graphs when solving a problem like this, they can use other strategies). Then solve the related real-world inequalities, ie which plumber is cheapest for short jobs, which plumber is cheapest for long jobs	They solve problems involving simultaneous linear equations and linear inequalities in 2 variables graphically and algebraically and justify solutions. For example, students can: - Use the elimination OR substitution method to algebraically solve pairs of linear simultaneous equations and verify that the solution works in both equations (where equations are presented with variables in the same order and on the same side of the equals sign for this standard of work). This includes: - Having to adjust one or both equations to ensure that one variable will be eliminated when using the elimination method - Having to use the substitution method when x and y both have coefficients which are not 1. - Use graphing software (eg Desmos or GeoGebra) to confirm that the intersection point of the graphs of the simultaneous equations matches the solution that was obtained algebraically. - Translate worded problems into pairs of simultaneous equations (which end up having both variables in the same order and on the same side of the equals	They solve problems involving simultaneous linear equations and linear inequalities in 2 variables graphically and algebraically and justify solutions. They can use more than one algebraic method, identify the most efficient method for the situation and use freehand graph sketching to double check solutions. For example, students can: - Use the elimination AND substitution method to algebraically solve pairs of linear simultaneous equations and verify that the solution works in both equations (where equations are presented with variables in different orders and on different sides of the equals sign for this standard of work). - Justify whether the elimination method or the substitution method would be more efficient for specific pairs of equations. - Use freehand sketching (transpose both equations into the form $y = mx + c$ and sketch them both) to confirm that the intersection point of the graphs of the simultaneous equations matches the solution that was obtained algebraically. - Translate worded problems into pairs of simultaneous equations (which end up having variables in different orders and on different sides of the equals sign for this

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	charge for each plumber. (Students don't necessarily have to write the algebra rule from the words to generate the graphs when solving a problem like this, they can use other strategies). – Solve the related real-world inequalities, ie which plumber is cheapest for short jobs, which plumber is cheapest for long jobs. • Follow a supplied process to: – Use graphing software (eg Desmos) to graphically solve linear inequalities for real-world problems where there are two variables and a fixed number constraint (eg identify all the combinations of trips to the movies (costing \$12) and ice-skating sessions (costing \$19) that would cost less than \$150 by entering $12x + 19y < 150$ into Desmos and then identifying some coordinate pairs from the graph which make the inequality true).	and what number of hours is the point to change plumbers? • Use graphing software (eg Desmos) to graphically solve linear inequalities for real-world problems where there are two variables and a fixed number constraint (eg identify all the combinations of trips to the movies (costing \$12) and ice-skating sessions (costing \$19) that would cost less than \$150 by entering $12x + 19y > 150$ into Desmos and then identifying the coordinate pairs from the graph which make the inequality true). • Interpret and communicate the solutions using language (and other representations) that match the original context of the problem.	sign), solve the equations then interpret and communicate the solutions using language (and other representations) that match the original context of the problem.	standard of work), solve the equations then interpret and communicate the solutions using language (and other representations) that match the original context of the problem.

Measurement

Students interpret and use logarithmic scales representing small or large quantities or change in applied contexts.

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Measurement	With support, students interpret and use logarithmic scales representing small or large quantities or change in applied contexts. For example, students can: - Read a logarithmic scale - Give examples of real-life situations in which logarithmic scales are used (eg, decibel scale, Richter scale). - Recognise that logarithmic scales are used because it is hard to represent numbers that cover a very large range on a linear scale when presented with a Richter scale presented on the usual logarithmic scale and then presented again on a linear scale	Students interpret and use logarithmic scales representing small or large quantities or change in applied contexts. For example, students can: - Read and interpret common logarithmic scales such as Richter scales and decibel scales and identify the magnitude of the change, ie each reading on the scale is 10 times bigger than the previous reading. (eg on the Richter scale, magnitude 6 is 10 times stronger than magnitude 5, magnitude 7 is 100 times stronger than 5. On the decibel scale, a change in power by a factor of 10 represents a 10 dB change in level, so 60 dB is 10 times more powerful than 50 dB and 70 dB is 100 times more powerful than 50 dB. Similarly a magnitude 6 earthquake releases one hundredth of the energy of a magnitude 8 event). - Students explain why we use logarithmic scales ie it is hard to represent numbers that cover a very large range on a linear scale and demonstrate this by showing the Richter or decibel scales represented on logarithmic scales and then again on a linear scales.		

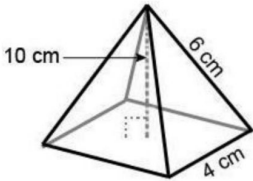
Students solve measurement problems involving surface area and volume of composite objects.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Measurement	With support, they solve measurement problems involving surface area and volume of composite objects. For example, students can: - Describe appropriate units for measuring surface area or volume. - Identify in a real-world problem if they should be determining the surface area or the volume. - Identify the different solids which comprise a composite object. - Match rectangular prisms, triangular prisms and cylinders to the appropriate formulae for volume. - Visualise a composite solid as its different exposed faces to find surface area.	They solve measurement problems involving surface area and volume of composite objects. For example, students can: - Select the correct formulas from a list of surface area and volume formulas for 3D objects, and use them to solve surface area and volume problems involving composite solids formed from any 2 of rectangular prism, triangular prism or cylinder, where necessary measurements are included in the diagram, eg - Rectangular prism joined to rectangular prism - Rectangular prism joined to triangular prism. - Rectangular prism joined to the end of a cylinder. - Cylinder joined to another cylinder at the ends.	They solve measurement problems involving surface area and volume of composite objects where diagrams need to be constructed for composite solids made from prisms and cylinders and where pyramids, cones, semi-cylinders and hemispheres are included in supplied diagrams. For example, students can: - Interpret a worded situation to make a diagram of a composite solid that consists of any two of: rectangular prism, triangular prism or cylinder, selecting the correct formulas from a list of surface area and volume formulas for 3D objects. eg, you are given two options for a two-tiered wedding cake. Both options have tiers with a height of 8 cm. On both cakes the top tier is two thirds the width of the bottom tier. In the cake with cylindrical tiers the width of the lower tier is 36 cm and in the cake with square prism tiers the lower tier is 30 cm wide. Which choice will have the most cake and by how much? Which choice will require the most icing and by how much? Justify your answers. - Solve surface area and volume problems involving composite solids made from 2 solids where rectangular pyramids, cones, semi-cylinders and hemispheres	They solve measurement problems involving surface area and volume of composite objects where: diagrams need to be constructed for a wider variety of composite solids, missing dimensions need to be worked out, composite solids are made from more than two regular solids. For example, students can: - Use the volume of a composite solid (made from 2 solids, one of which is either a rectangular pyramid or a cone) and other necessary measurements to work out a missing length and hence work out the surface area of the composite solid, selecting the correct formulas from a list of surface area and volume formulas for 3D objects. - Optimize the surface area of a composite solid made from a square pyramid joined to a square prism or a cone joined to a cylinder when given a volume. Students select five different sets of dimensions to justify that they have found the solid with the minimum surface area or students might generate a more complete data set if using technology (eg Excel). - Use Pythagoras' Theorem to work out the height or slant height of a rectangular pyramid or cone and hence determine the volume or surface area of the composite solid that includes the pyramid or cone.

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			are included in the composite solids and where necessary measurements are included in the diagram, eg – Rectangular prism joined to rectangular pyramid – Rectangular prism joined to semi-cylinder – Cylinder joined to cone – Cylinder joined to hemisphere. (Composite objects involving curved surfaces are joined at flat faces)	• Construct correct sketches of composite solids consisting of 2 solids (one of which is either a rectangular pyramid or a cone) from worded problems in order to find volume and surface area. • Find the surface area and the volume for composite solids made from any 3 different solids where necessary measurements are included in the diagram. (Composite objects involving objects with curved faces are joined at flat faces)

Students apply Pythagoras' theorem and trigonometry to solve practical problems involving right-angled triangles.

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Measurement	With support, students apply Pythagoras' theorem and trigonometry to solve practical problems involving right-angled triangles. For example, students can: • Find unknown side lengths and angles in straightforward 2D problems involving application and understanding of: Pythagoras' theorem and trigonometry where: – a diagram is provided and support given to ensure students select and apply relevant formulas – the meanings of angle of elevation, angle of depression	Students apply Pythagoras' theorem and trigonometry to solve practical problems involving right-angled triangles. For example, students can: • Find unknown side lengths and angles in straightforward 2D worded problems using Pythagoras' Theorem or trigonometry in conjunction with angles of elevation, angles of depression or bearings to find a solution where: – a 2D diagram is established from a worded situation – the relevant formula is correctly selected from a supplied list that also includes irrelevant formulae.	Students apply Pythagoras' theorem and trigonometry to solve practical problems involving right-angled triangles where necessary information needs to be deduced from supplied information and some situations involve supplied diagrams of 3D objects. For example, student can: • Find unknown side lengths and angles in 2D and 3D problems using Pythagoras' Theorem or trigonometry in conjunction with angles of elevation, angles of depression or bearings to find a solution where: – A right-angle triangle is established from a worded problem or diagram scenario that might require some	Students apply Pythagoras' theorem and trigonometry to solve practical problems involving right-angled triangles where diagrams need to be drawn for 2D and 3D situations and contexts are made more complicated through different methods for writing bearings, multiple triangles or unit conversions. For example, students can: • Find unknown side lengths and angles in 2D and 3D problems using Pythagoras' Theorem or trigonometry in conjunction with at least any two of: angles of elevation, angles of depression, true bearings (eg 115°T), compass bearings (eg $S65^\circ\text{E}$), multiple triangles, conversion of units. Triangle/s are established from real-world worded 2D and 3D situations.

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	or bearings are explained with a similar example.	Examples:	information to be inferred or deduced from provided information and questions may include some unnecessary information as a distractor. Examples: – A guide rope with an angle of elevation of 23° is fastened to the ground and 2.5 m below the top of an 11 m high flagpole. Find the distance between the base of the pole and the point on the ground where the guide rope is fastened, as well as the length of the guide rope. – Find the angle of elevation and length of the slant height (that reaches from the midpoint of the square side to the apex) in this square based pyramid.  – I am 178 cm tall and my eyes are 7.5 cm lower than the top of my head. I am looking up at an angle of elevation of 47° to the top of a tree that I am standing 5.25 m from. How high is the tree? What assumptions have you made? – Liam cycles at 24 km/h for 10 minutes on a bearing $S46^\circ W$. He then cycles due north until he is due west of his starting point. How far is he from his starting point?	Examples: – My plane leaves the airport at 8 am for a scenic flight and flies 96 km on a bearing of $062^\circ T$. I then turn clockwise 90° and fly for a further 40 km. From this point I fly back to the airport, how far will the total journey be and what compass bearing will I need to fly on for the final leg? How long will the entire journey take if the plane travels an average speed of 179 km/h (assuming I travel at a constant speed). – A lighthouse 20 m tall is on the top of cliffs 180 m high. At 9 am the angle of depression from the observation deck on top of the lighthouse to a 12 m long ship was $20^\circ 10'$ and at 9:10 am it was $9^\circ 35'$. Find the average speed of the ship. – From my starting point I walked 5.2 km on a bearing of $S70^\circ E$. I then walked a further 3.4 km on a bearing of $S20^\circ E$. If I then walk directly back to my starting point, what true bearing will I need to walk on? How far will I have walked in total once I arrive back at my starting point? – A square based pyramid has a diagonal measurement of 14.4 cm on the base and a height of 132 mm. Find the angle of elevation the slant height makes with the base of the pyramid.

Students identify the impact of measurement errors on the accuracy of results.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Measurement	With support, they identify the impact of measurement errors on the accuracy of results For example, students can: - Recognise that a 1 cent/hour error in pay rates has far more impact over a large number of employees than over a small number of employees, eg compare the effect of this error over 10 employees versus 1 000 employees over a year. - Recognise that a 1 cent/hour error in pay rates has far more impact over a large period of time than over a small period of time, eg compare the effect of this error for one employee over one week versus the whole year versus 10 years.	They identify the impact of measurement errors on the accuracy of results For example, students can: - Identify the impact of inaccurate measurements in compounding calculations where lengths are used to calculate volumes in practical contexts, eg demonstrate the difference in volume when an intended 10 cm x 5 cm x 4 cm rectangular prism (eg a Fruit Box) is measured inaccurately as 9.8 cm x 4.8 cm x 3.8 cm. - Identify the impact of errors in pay in compounding calculations, eg work out the overall cost to a company of a 1 cent/hour error in pay rates over a large number of employees and over a long period of time, eg 1 000 employees over 20 years. - Identify the impact of inaccurate measurements in compounding calculations where lengths are used to calculate areas in practical contexts, eg how many tiles under or over the correct supply would result from inaccurately measuring the length and width of a 10 m x 5 m floor as 9.9 m x 4.9 m or as 10.1 m x 5.1 m if there are 16 tiles per square metre? How much under supply or over supply would result from making this error on ten of the same sized floors?	They identify the impact of measurement errors on the accuracy of results by showing the impact of errors in angles on distance and translating measurement errors into costs. For example, students can: - Explain sources of error in more involved situations, such as the impact of measuring in degrees-minutes-seconds rather than degrees in large scale calculations, such as using trigonometry to calculate large distances for navigation. - Evaluate the impact of not measuring accurately, by demonstrating with different values and translating a measurement error into an actual value, eg - Incorrectly measure the floor and buy too many tiles and translate to a dollar value (including researching the price of the tiles). - Incorrectly measure the length, width and height of a concrete slab for a shed and buy too much concrete and translate the error to a dollar value (including researching the price of the concrete). - Justify why and where it may be useful to under- and over-estimate measurements.	

Students use mathematical modelling to solve practical problems involving proportion and scaling, evaluating and modifying models, and reporting assumptions, methods and findings.

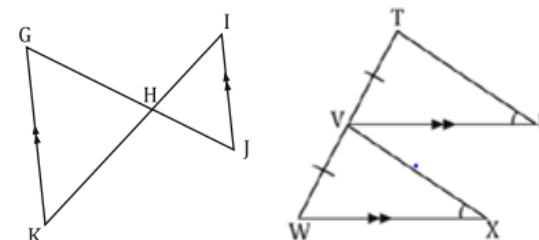
Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Measurement	With support, students use mathematical modelling to solve simple practical problems involving proportion and scaling, evaluating and modifying models, and reporting assumptions, methods and findings. For example, students can: - With support use a mathematical modelling approach that allows them to: - Translate a real-world problem into maths - Use maths to solve the problem - Interpret and communicate the solution back in the real-world context with some - justification - evaluation and modification - identification of assumptions - Given a house plan, increase the wall lengths (maintaining the height) by a whole number scale factors (eg 2 x longer, 3 x longer) and evaluate the impact upon the costs of external wall cladding.	Students use mathematical modelling to solve practical problems involving proportion and scaling, evaluating and modifying models, and reporting assumptions, methods and findings. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: - Translating the problem into maths - Using maths to solve the problem - Interpreting and communicating the solution back in the real-world context with: - justification - evaluation and modification - reporting of assumptions, methods and findings For example, students can: - Given a house plan, confirm that the plan is in proportion with the real-world measurements. Increase the wall lengths (maintaining the height) by a scale factor (eg 20% increase or increase by a scale factor of 1.2) and evaluate the impact upon the costs of: - External wall cladding. - Floor coverings.	Students use mathematical modelling to solve practical problems involving proportion and scaling, evaluating, modifying and refining models, and reporting assumptions, methods and findings, including the outcomes of the refined models. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: - Translating the problem into maths - Using maths to solve the problem - Interpreting and communicating the solution back in the real-world context with: - justification - evaluation, modification and refinements - reporting of assumptions, methods and findings - Given a freehand sketch of a house plan, convert it to a scale diagram. - Refine the plan of the house to meet a given requirement, eg - Adjusting the lengths and widths on the plan to meet a certain budget for the wall cladding. - Adjusting the lengths and widths on the plan to meet a certain budget for the floor covering.	Students use mathematical modelling to solve practical problems involving proportion and scaling, evaluating and modifying models and refining models, and reporting assumptions, methods and findings, including the reasons to justify the refinements, and making and testing hypotheses related to the original problem. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: - Translating the problem into maths - Using maths to solve the problem - Interpreting and communicating the solution back in the real-world context with: - justification - evaluation, modification and refinements - hypotheses (posed and tested) related to the original problem - reporting of assumptions, methods and findings For example, students can: - Re-draw a house plan with proportional lengths and widths when the original plan is not in proportion with the real-world measurements.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
				- Independently hypothesise about the impact on the cost of external wall cladding when increasing the length and width of the house by different scale factors (eg length increases by scale factor of 1.2 and width increases by scale factor of 1.35 so that the house plan suits the shape of the block of land). - Independently hypothesise about the impact on the cost of floor covering when increasing the length and width of the house by different scale factors (eg length increases by scale factor of 1.2 and width increases by scale factor of 1.35 so that the house plan suits the shape of the block of land). - Independently hypothesises about the impact of adding or removing a bedroom on the cost of wall cladding and floor coverings and tests their hypotheses providing thorough justification of their findings.

Space

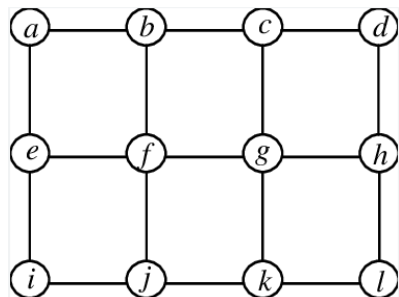
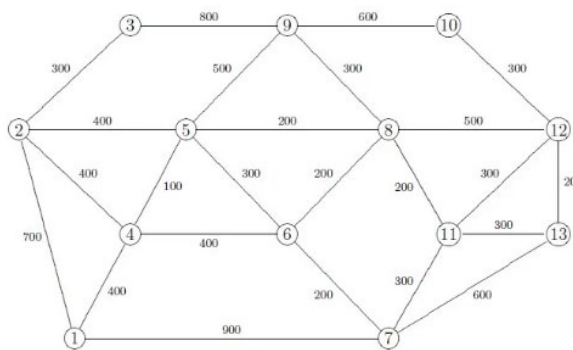
Students use deductive reasoning, theorems and algorithms to solve spatial problems

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Space	With support, they use deductive reasoning, theorems and algorithms to solve spatial problems For example, students can: - Follow given pseudocode to produce a given shape. Eg, square or rectangle - With support, give real life examples of similarity and congruence. For example, students can: - Explain that A3, A4 and A5 paper are similar. - Explain that scaling Pythagorean triples will produce similar triangles. - Given side lengths of a triangle can use a scale factor to determine dimension of another similar triangle.	They use deductive reasoning, theorems and algorithms to solve spatial problems. For example, students can: - Design pseudocode (a set of ordered instructions) that will produce a specified shape, for example: an equilateral triangle of given side length. - Understand the difference between a repeated practical verification and a proof. - Provide a justification based on logical reasoning that two triangles are congruent or similar. - Solve problems that require students to identify similarity or congruence and apply this understanding to find unknown values. Represent this using a sequence of logically connected statements. For example: - Prove that $\triangle ABC$ is similar to $\triangle DEF$ (Using either AA, SAS or SSS) - If $A = 80^\circ$, $F = 55^\circ$, Deduce the angle at E. - Deduce the value of x	They use deductive reasoning, theorems and algorithms to solve multi-step spatial problems. For example, students can: - Design pseudocode that will produce a given pattern, then use a range of test cases to confirm their output. Eg give a set of detailed instructions that will produce a series of similar triangles. - Solve problems that require identifying similarity or congruence and apply this understanding to find unknown values. Problems will involve multiple steps drawing on geometric theorems such as angle sum of triangle/quadrilateral, vertically opposite, adjacent, corresponding, co-interior angles and alternate angles to find missing angles or side lengths, utilising knowledge of similarity and congruence tests. For example: - Determine if these triangles are similar (or not) using similarity tests and other geometric theorems.	They use deductive reasoning, theorems and algorithms to solve multi-step spatial problems which require formal proof notation. For example, students can: - Design pseudocode that will produce a given pattern within specified parameters, then use a range of test cases to confirm their output. Eg, give a set of detailed instructions that will produce a series of similar triangles that all have a side length within a given range and a specified angle. - Students will use formal notation to develop and communicate proofs. Proofs will involve multiple steps drawing on geometric theorems such as angle sum of triangle/quadrilateral, vertically opposite, adjacent, corresponding, co-interior angles and alternate angles to find missing angles or side lengths, utilising knowledge of similarity and congruence tests. For example: - Develop a proof that shows that two triangles are similar (or not) using notation to communicate your reasoning.



Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
		Establish if these pairs of triangles are similar or congruent, by stating congruent and similar features.	For example: List all of the congruent features using correct notation and then state whether the following pairs of triangles are congruent and by which congruence tests (SSS, SAS, ASA, AAS or HL/RHS.)	

Students interpret networks used to represent practical situations and describe connectedness.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Space	With support, students interpret networks used to represent practical situations and describe connectedness. For example, students can: - Identify vertices/nodes and edges on a network diagram. - Find alternative routes on a virtual map between two given places. - Find a route that would pass through 4 places, developing a logical route. For example, given 4 locations in the CBD plan a sensible walking route that would minimize walking distance.	Students interpret networks used to represent practical situations and describe connectedness. For example, students can: - Interpret familiar networks including interpreting the meaning of vertices/nodes and edges on real-world network diagrams. Answer questions relating to connectedness of nodes. eg: - Network diagram of road routes with towns (vertices/nodes) and roads (edges) - Network diagram of train routes with train stations (vertices/nodes) and tracks (edges) - Network diagram of airline routes with airports (vertices/nodes) and flight paths (edges) - Network diagram of sewer system with houses and the treatment plant (vertices/nodes) and pipes (edges) - Network diagram of social relationships with people (vertices/nodes) and friendships (edges) - Network diagram of internet cabling in a school	Students systematically analyse networks used to represent practical situations and describe connectedness. For example, students can: - Compare networks by systematically analysing possible paths for a network that has multiple routes given some constraints. eg. list all possible pathways to get from <i>a</i> to <i>l</i> (which letters stay in the order they occur in the alphabet) eg. <i>a, e, i, j, k, l</i>  A 3-row by 4-column grid structure composed of 12 nodes. Download Scientific Diagram (researchgate.net) - Evaluate the connectedness of real-world situations. For example, if there were 5 people in a group and everyone shook hands with everyone else, how many handshakes were there? What happens if a 6th person joins the group? Develop a rule for any of number of people.	Students systematically analyse more complex networks used to represent practical situations and describe connectedness. For example, students can: - Determine possible paths for a network that has multiple routes and evaluate routes where edges may have weights or nodes vary in their connectedness and are not arranged in a regular pattern. Eg: determine the shortest route between 1 and 13, and justify your response.  Network maths problem Physics Forums - Eg suggest a route that would allow each vertex/node and/or edge to be visited once only. Eg a postman travelling along each street once and returning to their starting point. - Analyse the degrees of connectedness in your school's fire evacuation plan as an existing network and make suggestions to improve efficiency or justify why the current plan is best.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
		– Suggest different routes on a virtual map that allow the same nodes to be visited using different edges.		

Statistics

They plan and conduct statistical investigations involving bivariate data.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Statistics	With support, they plan and conduct statistical investigations involving bivariate data. For example, students can: - Follow a process supplied by the teacher to collect reliable primary bivariate data. - Use a secondary source nominated by the teacher to collect secondary bivariate data. - Follow guidance from the teacher to draw conclusions from data.	They plan and conduct statistical investigations involving bivariate data. For example, students can: - Plan to collect data that is relevant to the question(s) being investigated. - When planning and conducting a statistical investigation involving bivariate data, they evaluate a sampling method designed by themselves or someone else to collect primary data, identify factors which could lead to unreliable and unfair data, (eg sample is too small, sample is not random, only one sample was taken instead of comparing results from multiple random samples of adequate size) and improve the sampling method so that it produces fair and reliable data. - When planning and conducting a statistical investigation involving secondary bivariate data, they evaluate whether the data is reliable by naming the source of the data and commenting on whether they think the source can be trusted, giving valid reasons why. - Draw valid conclusions which match the evidence that was collected and provide some valid reasons.	They plan and conduct statistical investigations involving bivariate data by checking the reliability of secondary data sources and identifying the limitations of inferences. For example, students can: - Explain why a usually reliable secondary source of data might have specific data sets that are not trustworthy (eg a data set is missing because a survey wasn't done due to COVID; it is hard to collect accurate data in some circumstances, eg Census data was collected twice for some people (in the house where they normally live and in the house where they were staying on census night); a worker in a data collection agency has produced a data set without following normal protocols for ensuring data reliability; a worker who was tired at the end of the day accidentally posted the wrong data set) and explains the steps they would take to try and ensure reliable data (eg comparing data sets from different usually reliable sources). - Search a secondary source of bivariate data on the internet to find out how the data was collected and comment on how easy it is to identify the data	They plan and conduct statistical investigations involving bivariate data by checking reliable secondary data sources to see if they are truly independent and planning further investigations to remove the limitations on inferences. For example, students can: - Draw valid conclusions which match the evidence that was collected, identify limitations of inferences giving valid reasons then justify whether the limitations can be removed or not through further investigation. If the limitations can be removed, the student provides a plan for further investigation which would remove the current limitations on the inferences. (The student might carry out this further investigation if there is time and if it is practicable). - Search two (or more) different sources of secondary bivariate data on the internet (that are reporting on the same issue) to find out whether they are relying on the same source or whether they are truly independent sources. Describe the evidence that confirms whether the data sources are independent or not and describe the attempted confirmation process if independence of data sets cannot be confirmed. Comment on how much trust they would put in each data

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
			collection method and then comment on how much trust they would put in the data, giving valid reasons. Draw valid conclusions which match the evidence that was collected, provide detailed reasons and identify limitations of inferences giving valid justifications.	set giving valid reasons including citing strengths and weaknesses.

Students represent the distribution of data involving 2 variables, using tables and scatter plots, and comment on possible association.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Statistics	With support, students represent the distribution of data involving 2 variables, using tables and scatter plots, and comment on possible association. For example, students can: - Follow a process supplied by the teacher to construct scatter plots with accurate axes and draw valid conclusions about the direction and strength of correlation between two numerical variables, ie - Direction – positive, negative or no correlation - Strength – linear, strong, moderate, weak, no correlation. - Follow a process supplied by the teacher to draw a line of best fit by eye through a scatter plot where linear or strong correlation is indicated and use the line of best fit to work out values for data points	Students represent the distribution of data involving 2 variables, using tables and scatter plots, and comment on possible association. For example, students can: - Construct scatter plots with accurate axes and draw valid conclusions about the direction and strength of correlation between two numerical variables, ie - Direction – positive, negative or no correlation - Strength – linear, strong, moderate, weak, no correlation. - Draw a line of best fit by eye through a scatter plot where linear or strong correlation is indicated and use the line of best fit to work out values for data points that were not in the original data set. - Use a technological tool (eg Excel or Desmos) to find a line of best fit	Students represent the distribution of data involving 2 variables, using tables and scatter plots, comment on possible association, and evaluate a generated line of best fit. For example, students can: Consistently construct tables showing the real-world data points and the corresponding data points which are predicted from the line of best fit lined up with each other, to enable easy comparison, and then evaluate whether the line of best fit is predicting data values which are close enough to the real-world data over the whole data range. This might include students stipulating part of the data range where the line of best fit seems accurate enough and other parts of the data range where the line of best does not seem to be adequate.	Students represent the distribution of data involving 2 variables, using tables and scatter plots, comment on possible association and use evidence of comparison by changing parameters to evaluate generated lines of best fit. For example, students can: - Use a technological tool (eg Excel or Desmos) to justify a choice of line of best fit, using evidence of comparison, by changing each of the coefficients and constants in the algebra rule one at a time (at least once each) and showing that the new rules do not fit the data as well as the original model, eg - For linear rules of best fit, students show that the values of m and c they have chosen in their rule of the form $y = mx + c$ are the best possible values by changing m then c in the technological

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	that were not in the original data set. Follow a process supplied by the teacher to construct two-way tables and draw valid conclusions about the relationship between categorical variables, eg investigating the relationship between the grade of a student at school and the sport they most prefer.	for a scatter plot, where linear or strong correlation is indicated, by selecting the most appropriate type of correlation (linear, polynomial or exponential), use the generated algebra rule to work out values for points that were not in the original data set and then evaluate whether the line of best fit is predicting values which seem to be close enough to the real-world data. - Construct two-way tables and draw valid conclusions about the relationship between categorical variables, eg investigating the relationship between the grade of a student at school and the sport they most prefer. - Understand that, in general, correlation on a graph does not prove causation and, in specific cases, decide whether correlation might be grounds for investigating causation or whether it is coincidental, giving valid reasons why.		tool and showing that the changed models do not fit the actual real-world data as well as the original model. Similarly for quadratic or exponential rules in the forms of $y = ax^2 + bx + c$ and $y = ab^x$ are the best possible by following the process described above.

They analyse inferences and conclusions in the media, noting potential sources of bias.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Statistics	With support, they analyse inferences and conclusions in the media, noting potential sources of bias. For example, students can: Follow a process supplied by the teacher to check the specific listed items in the at standard box, when prompted by the teacher about	They analyse inferences and conclusions in the media, noting potential sources of bias. For example, students can: Accurately summarise conclusions from media reports using their own words and other visual	They thoroughly analyse inferences and conclusions in the media, noting potential sources of bias and suggest and justify modifications to address said bias. For example, students can:	They analyse inferences and conclusions in the media thoroughly, noting potential sources of bias then re-write a media report, incorporating all necessary improvements, so that it is clear, concise, thorough and unbiased and describe reasons for the unreliability or bias in the original report.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	each individual item, and decide whether the report is adequate with respect to each item when supported by the teacher in making these decisions. Can identify a small number of the factors in the At Standard list.	representations, eg tables and graphs. - Check the media report for factors that would increase or decrease confidence in the conclusions by using the list provided below. Factors that would increase confidence are: - Data being collected is directly related to the question being investigated - Samples were random, of adequate size and representative of the population - Multiple samples were used - Permissions were obtained for collection and use of the data - Processes are in place to ensure anonymity for people participating in a survey - Data is to be used for benefit, not harm - Secondary data is from sources that may have been trustworthy in the past - Graph scales start at 0 - Graph scales have equal increments (it doesn't have to be the same increment on each axis but the increments on each axis should be equal) - Tables and graphs are presented fairly and not trying to emphasize just some of the data - The stated conclusions in the report match the evidence shown by the data	- Suggest improvements to remove concerns over bias and ethics with respect to: - Data collection - Data representation - Conclusions in media investigations. - Give valid reasons for each improvement that is suggested. - Display thoroughness in their analysis by checking most if not all, dot points in the list in the At Standard box and deciding whether the report is adequate or whether improvement is required with respect to each listed item.	For example, students can: Re-write a media report that contains unreliable or biased data or conclusions. Ensure that the re-written report contains: data, representations and conclusions which are: clear, concise, thorough and unbiased. Explain the changes made to the original media report and justify each change. Describe the specific influences and motivations which might have caused the original media report to be unreliable or biased.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
		– The conclusions agree with conclusions from other fair and ethical surveys on the same topic. • Students check more than half of these items when evaluating the media report. • Decide whether the conclusions are reliable and provide a summary of valid reasons to support their decision but also note potential sources of bias.		

Students compare the distribution of continuous numerical data, using various displays, and discuss distributions in terms of centre, spread, shape and outliers.

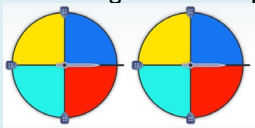
Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Statistics	With support, students compare the distribution of continuous numerical data, using various displays, and discuss distributions in terms of centre, spread, shape and outliers. For example, students can: • Construct two parallel box plots to compare two sets of data by following a process given by the teacher. • Describe the shapes of data distributions (ie symmetrical, positively skewed, negatively skewed, bimodal) when given examples of data sets with these different shapes to enable easy comparison. • Follow a supplied process to determine the mean, median, range and inter-quartile range for data sets.	Students compare the distribution of continuous numerical data, using various displays, and discuss distributions in terms of centre, spread, shape and outliers. For example, students can: • Construct two parallel box plots to compare two sets of data. • Compare two data sets by using more than one piece of evidence, ie – Constructing appropriate, easily comparable graphs aligned vertically or horizontally (eg histograms OR divided bar graphs with grouped data and relative frequencies, OR boxplots) – Describing the shapes of the data distributions (ie symmetrical, positively skewed, negatively skewed, bimodal)	Students compare the distribution of continuous numerical data, using various displays, and discuss distributions in terms of centre, spread, shape and outliers by comparing more than two data sets using comprehensive evidence and providing detailed reasons. For example, students can: • Compare more than two data sets using comprehensive evidence by: – Constructing appropriate, easily comparable graphs aligned vertically or horizontally (eg histograms OR divided bar graphs with grouped data and relative frequencies, OR boxplots) – Describing the shapes of the data distributions (ie	Students compare the distribution of continuous numerical data, using various displays, and discuss distributions in terms of centre, spread, shape and outliers by considering the relative strength of different pieces of evidence when comparing more than two data sets and designing further investigations to provide stronger evidence. For example, students can: • Demonstrate how different graph types for the same data set reveal different features of the data sets and independently use more than one graph type (eg histograms AND divided bar graphs with grouped data and relative frequencies AND boxplots) in conjunction with means, medians, ranges and inter-quartile ranges to draw more detailed conclusions when

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	• Draw conclusions which match the data when prompted. • Provide some valid reasons for conclusions when prompted.	– Summarising the centre and spread of data sets using mean, median, range and inter-quartile range – Comparing some aspects of the organised and summarised data (ie use some of table, graphs, means, medians, ranges and inter-quartile ranges) rather than just relying on one aspect and deciding whether there is a significant difference between the data sets and whether one set of data is more predictable (ie has less spread) than the other. – Drawing some valid conclusions and providing valid reasons. • Demonstrate how the shape (symmetrical, positively skewed, negatively skewed, bimodal) of a data distribution affects the 'look' of a boxplot and match supplied boxplots with the appropriate data distribution shape.	symmetrical, positively skewed, negatively skewed, bimodal) – Summarising the centre and spread of data sets using mean, median, range and inter-quartile range – Comparing all aspects of the organised and summarised data (ie use the tables, graphs, means, medians, ranges and inter-quartile ranges) rather than just relying on one aspect and deciding whether there are significant differences between the data sets and whether some sets of data are more predictable (ie have less spread) than the others. – Drawing valid conclusions and providing detailed reasons.	comparing more than two data sets. • Cite the relative strength of different pieces of evidence when providing valid detailed reasons to support inferences. • Provide a thorough plan, with appropriate sampling, for further data collection to address weaknesses in the current evidence by generating new, relevant evidence that would lead to more definite conclusions. • Demonstrate how the same data summary can match multiple data sets of different sizes and different distributions to justify why the data summary alone is insufficient when drawing reliable conclusions. (Eg show how different sized class sets of data with different distributions can result in the same values for the five number summary, the mean, the range and the inter-quartile range).

Probability

Students apply conditional probability to solve problems involving compound events.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Probability	With support, they apply conditional probability to solve problems involving compound events. For example, students can: - Follow a process given by the teacher to construct a tree diagram, two-way table or organised list to analyse compound events consisting of two single events (with and without replacement) and hence determine probabilities as fractions then decimals and percentages. - Use examples given by the teacher when solving problems involving language like: “if then”, “given”, “knowing that” to compare conditional probability situations where some prior outcomes are already known. - Follow a process given by the teacher to construct a Venn diagram for a given set of information and hence determine probabilities as fractions then decimals and percentages.	They apply conditional probability to solve problems involving compound events. For example, students can: - Use tree diagrams or organised lists to determine probabilities when solving problems involving compound events. The compound events at this standard could be any combination of two single events (with and without replacement) and the single events could be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container. - Use language like: “if then”, “given”, “knowing that” to compare conditional probability situations where some prior outcomes are already known, eg - The chance of getting two heads when I toss two coins is $\frac{1}{4}$, however, <u>if</u> I already know that one of the coins has come up heads <u>then</u> the chance of getting the second head with the last toss is $\frac{1}{2}$ - The chance of getting two numbers above 4 when throwing two dice is $\frac{4}{36} = \frac{1}{9}$, however, <u>given</u> that one of the dice has already come up with a number above 4 then the chance of getting a second	They apply conditional probability to solve problems involving compound events by analysing compound events which consist of combinations of three single events and a variety of known prior outcomes. For example, students can: - Use tree diagrams or organised lists to determine probabilities when solving problems involving compound events. The compound events at this standard could be any combination of three single events (with and without replacement) and the single events could be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container. - Use language like: “if then”, “given”, “knowing that” to determine conditional probabilities for situations involving compound events where some prior outcomes are already known and the compound events consist of any combination of three single events, eg when tossing three dice and already knowing two of the numbers that have come up, or when flicking three spinners and already knowing one of the colours that has come up or when drawing balls out of a	They apply conditional probability to solve problems involving compound events by fluently converting between different representations to compare different compound events which consist of combinations of three (or more) single events and a variety of known prior outcomes. For example, students can: Fluently convert between different representations (ie re-write tree diagrams as organised lists of all possibilities and re-write organised lists of all possibilities as tree diagrams) to compare percentage probabilities for compound events in two different real-world situations (eg what is the probability of getting 1, 2 and 3 when rolling two dice and flicking a spinner if I already know that I have rolled a 2, versus picking three cards with numbers on them without replacement when I don't know any of the numbers that have been picked so far). The compound events at this standard could be any combination of three (or more) single events, with and without replacement and different numbers of prior outcomes can be specified to vary the conditional probability, eg I already know I have rolled a 2, or I already know I have rolled a 2 and a 3. (Single events could be:

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
		number above 4 with the last toss is $\frac{1}{3}$ – The chance of getting two reds when using these two spinners  – is $\frac{1}{16}$, however, <u>knowing that</u> one of the spinners has already come up red then the chance of getting one more red on the last spin is $\frac{1}{4}$ • Use Venn diagrams with three circles to determine conditional probabilities when solving problems involving compound events. (eg construct a Venn diagram to show the numbers of students participating in Tennis, Football or Cross Country and combinations of these physical activity classes then determine the probability that a student participates in Cross Country knowing that they already play Tennis and Football).	container without replacement and already knowing two of the colours that have come out.	tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container).

Students design and conduct simulations involving conditional probability, using digital tools.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Probability	With support, students design and conduct simulations involving conditional probability, using digital tools. For example, students can: - Follow a process given by the teacher to create a digital experiment or simulation for compound events comprised of two single events, with and without replacement. (Single events could be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of two single events). - Follow a process given by the teacher to check that the results from the digital experiment or simulation are in reasonable agreement with the results that are expected from theoretical analysis (tree-diagram, two-way table or organised list).	Students design and conduct simulations involving conditional probability, using digital tools. For example, students can: - Use digital tools to design and conduct experiments or simulations for compound events comprised of two single events, with and without replacement. (Single events could be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of two single events). - Compare the results from the digital experiment or simulation with the results that are expected from theoretical analysis (tree-diagram or organised list). - Use digital tools to design and conduct an experiment or simulation to compare probabilities of independent and dependent events and see if the results from the simulation agree with the theoretical analysis (using a tree diagram or organised list), eg design a simulation to compare rolling two dice with picking two numbers out of a bag without replacement (the numbers 1, 2, 3, 4, 5, 6 are in the bag to start with).	Students design and conduct simulations involving conditional probability, using digital tools for compound events which consist of combinations of three single events. For example, students can: - Use digital tools to design and conduct experiments or simulations for compound events comprised of three single events, with and without replacement. (Single events could be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of three single events). - Compare the results from the digital experiment or simulation with the results that are expected from theoretical analysis (tree-diagram or organised list).	Students design and conduct simulations involving conditional probability, using digital tools for compound events which consist of combinations of three (or more) single events and demonstrate that the same simulation can accurately represent different real-world scenarios. For example, students can: Understand that different real-world situations can be equivalent from a probability point of view. Demonstrate this understanding by showing that the same digital simulation generates results that match the theoretical probabilities of three different real-world situations where the different situations use different objects and different symbol systems (eg one situation uses numbers on dice, one situation uses letters on cards and one situation uses colours on spinners). The three real-world situations being compared must be compound events comprised of three (or more) single events, with and without replacement. (Single events could be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of three single events).